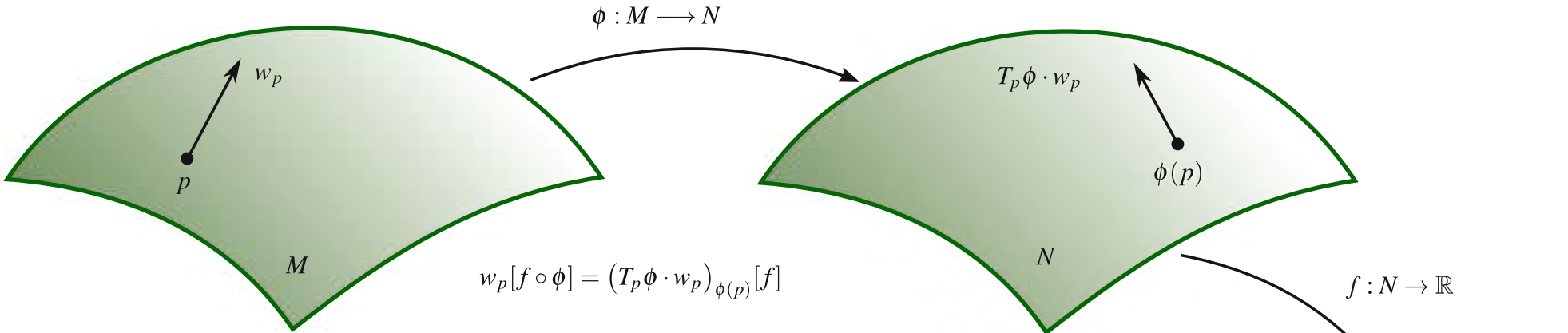


$$w_p[f \circ \phi] = \left(T_p\phi \cdot w_p\right)_{\phi(p)}[f]$$

$$\underbrace{w[f \circ \phi] \equiv w[T^*\phi \cdot f]}_{\text{function on } M} \qquad \underbrace{(T\phi \cdot w)[f]}_{\text{function on } N}$$

$$\underbrace{w[T^*\phi \cdot f] \stackrel{?}{=} (T\phi \cdot w)[f]}_{\text{functions on different manifolds}}$$

$$\underbrace{w[T^*\phi \cdot f] = ((T\phi \cdot w)[f]) \circ \phi \equiv T^*\phi \cdot ((T\phi \cdot w)[f])}_{\text{now both functions on } M}$$